

Lecture 6

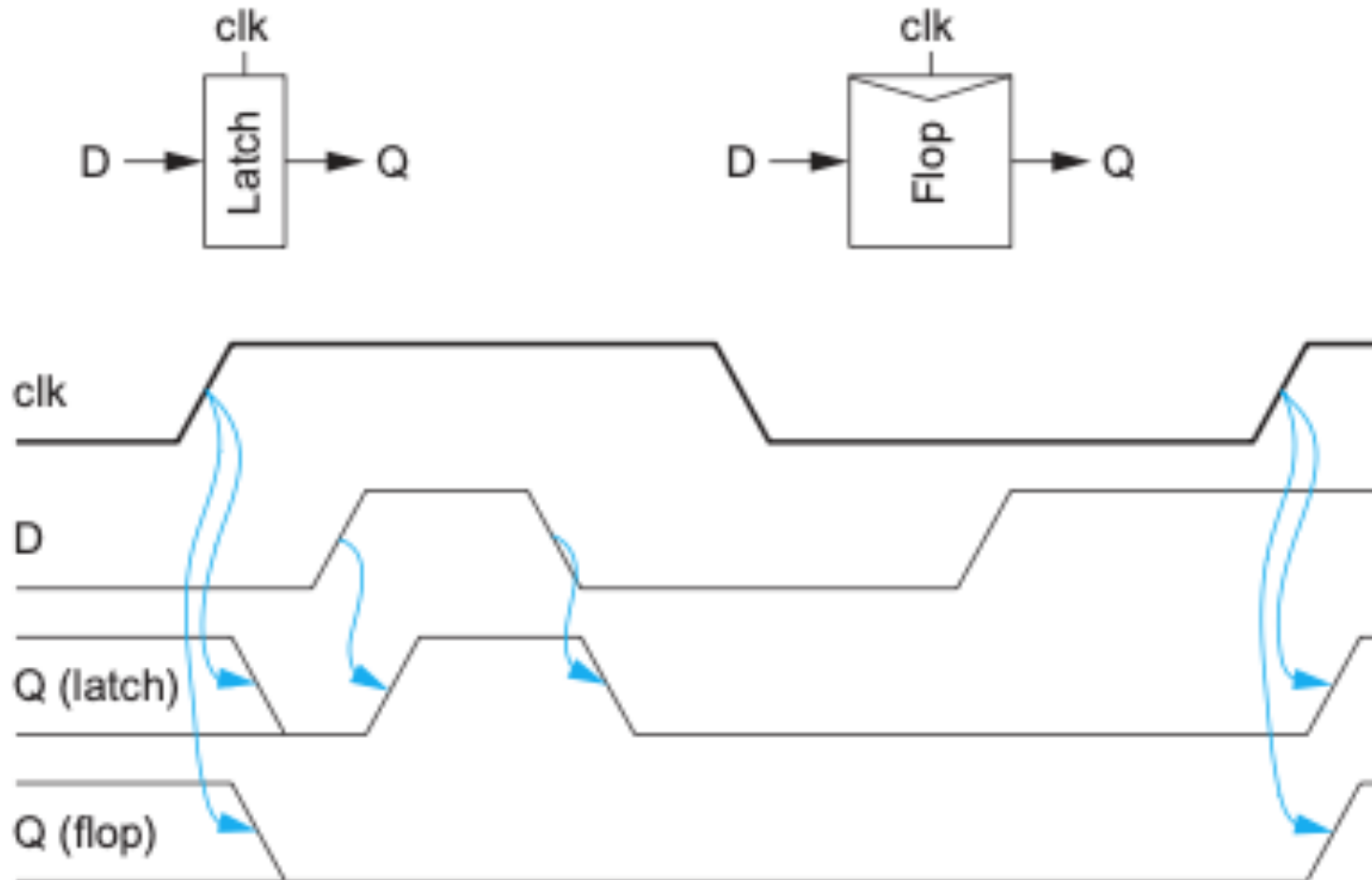
Sequential Circuits

Prof Peter YK Cheung

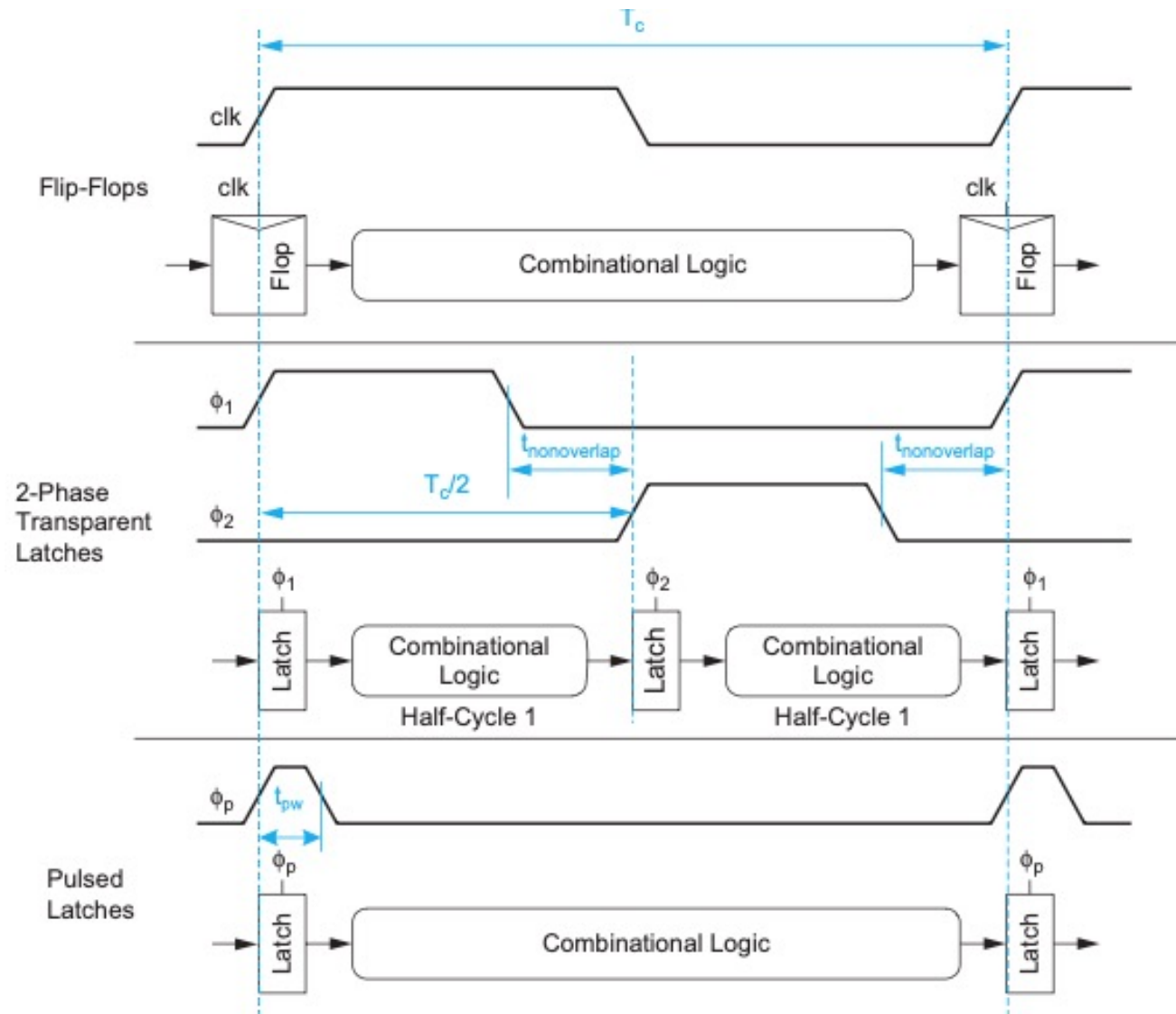
Department of Electrical & Electronic Engineering

URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/
E-mail: p.cheung@imperial.ac.uk

Latches vs Flip-flops



Clocking Styles

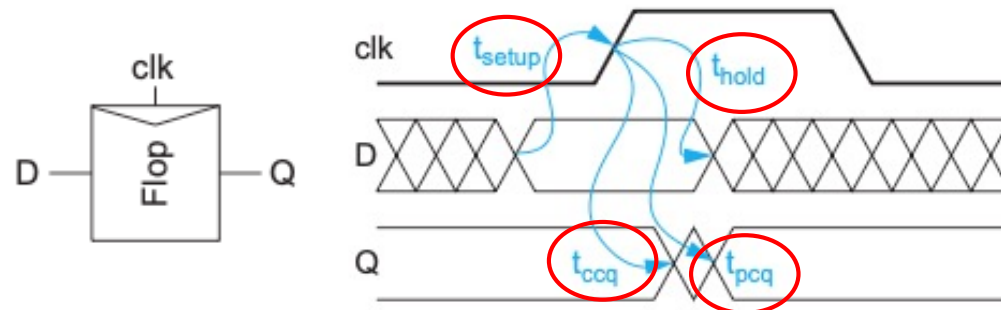


W&H p377-378

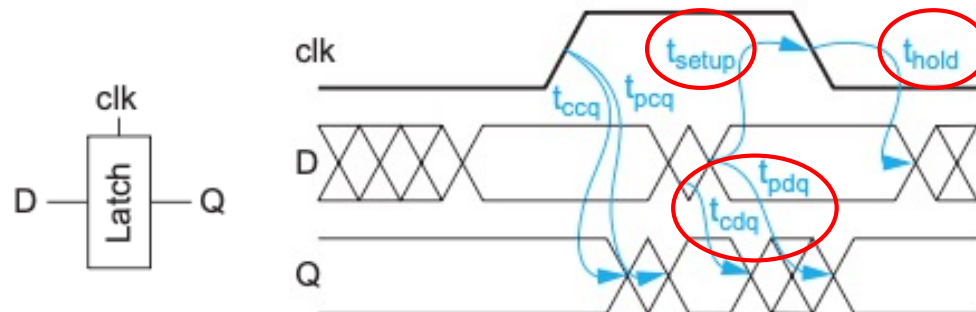
Timing considerations for sequential circuits



(a)



(b)

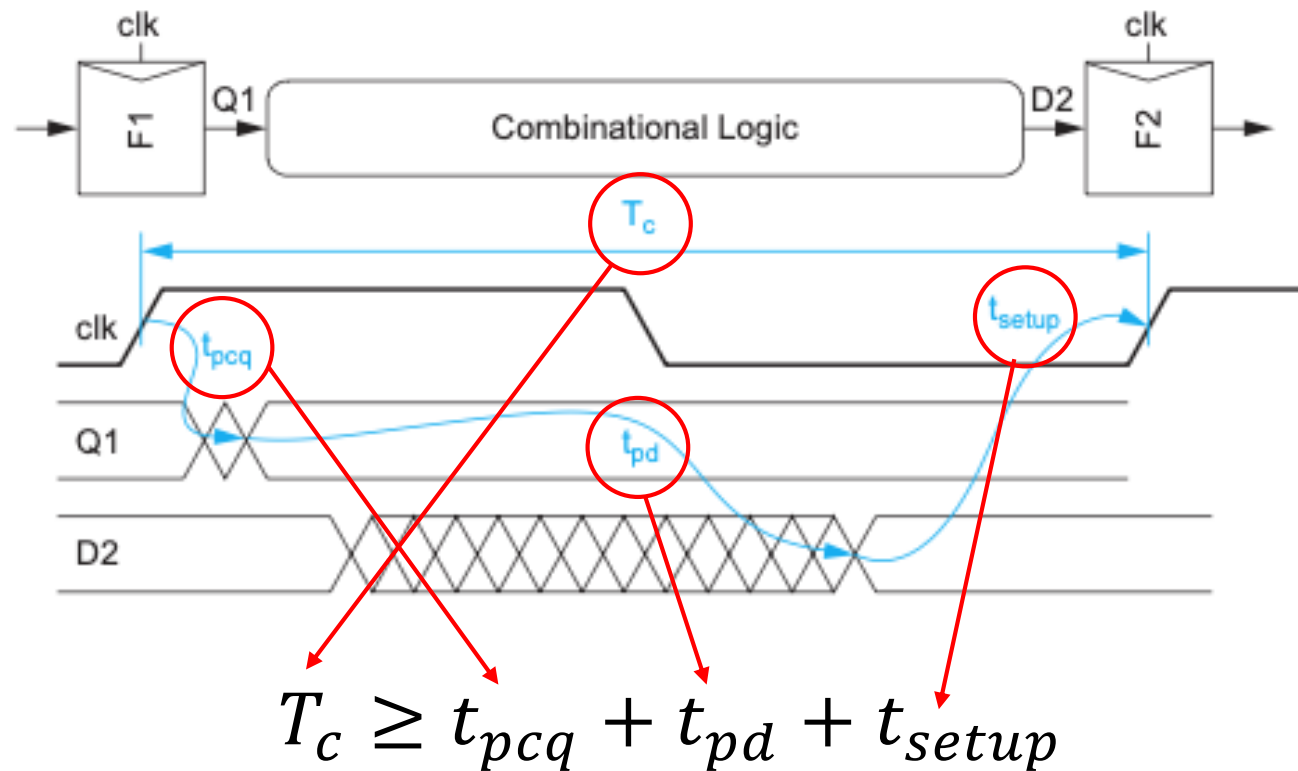


(c)

Term	Name
t_{pd}	Logic Propagation Delay
t_{cd}	Logic Contamination Delay
t_{pcq}	Latch/Flop Clock-to-Q Propagation Delay
t_{ccq}	Latch/Flop Clock-to-Q Contamination Delay
t_{pdq}	Latch D-to-Q Propagation Delay
t_{cdq}	Latch D-to-Q Contamination Delay
t_{setup}	Latch/Flop Setup Time
t_{hold}	Latch/Flop Hold Time

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Flip-flop Maximum Delay Constraint

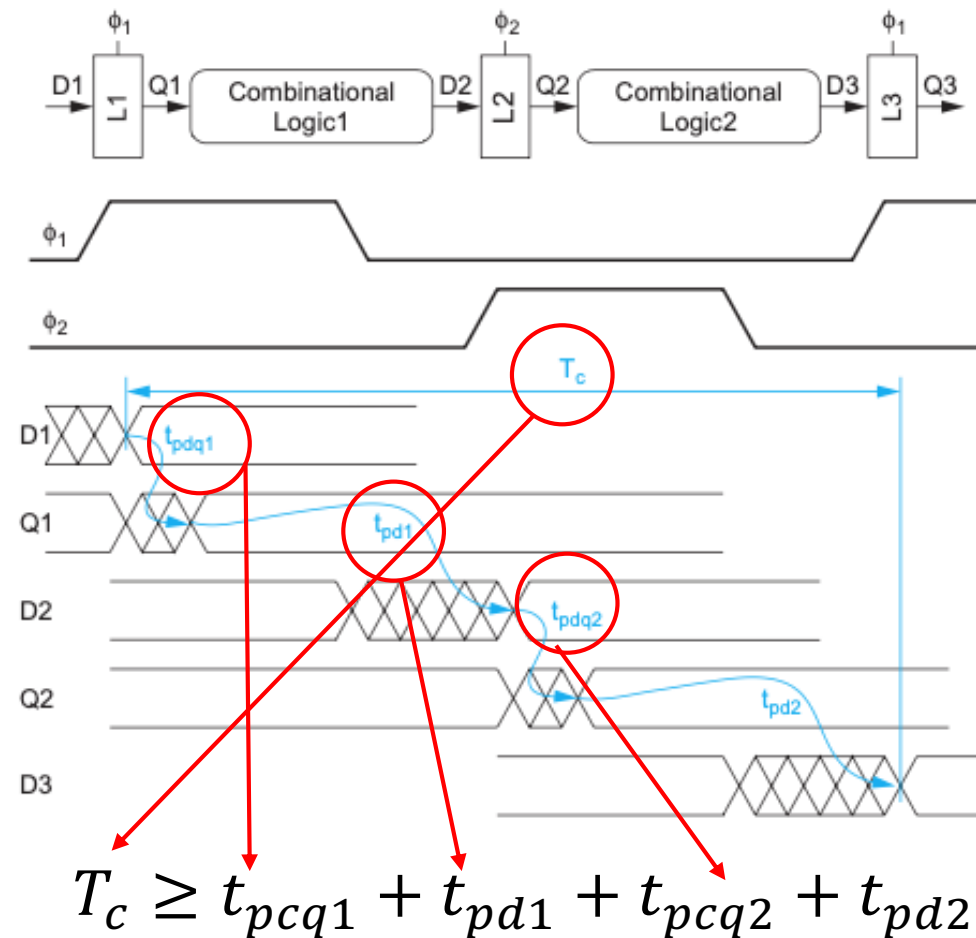


$$t_{pd} \leq T_c - (t_{setup} + t_{pcq})$$

sequencing overhead

W&H p380

Latch Maximum Delay Constraint

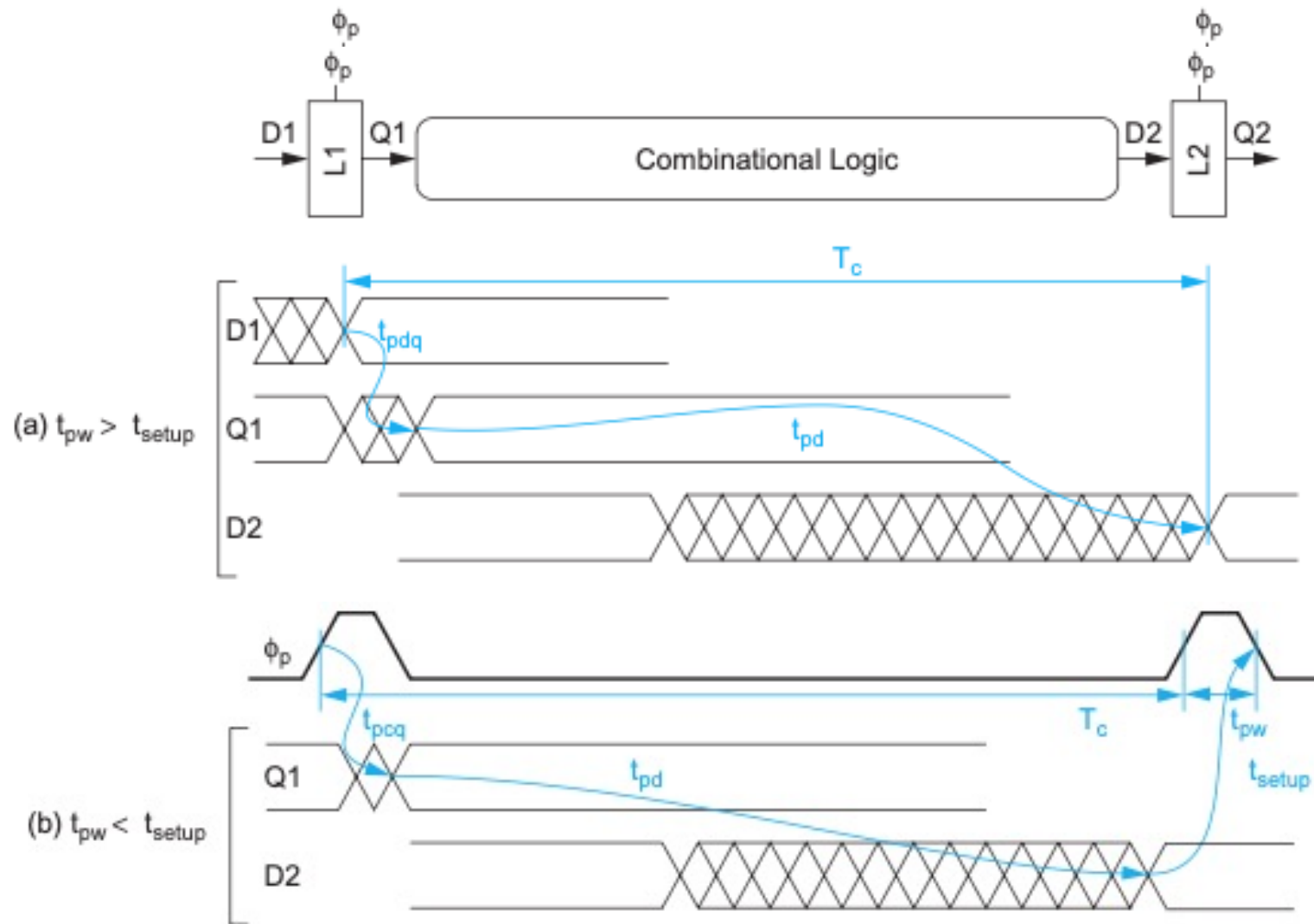


$$t_{pd} \leq T_c - (2t_{pcq})$$

sequencing overhead

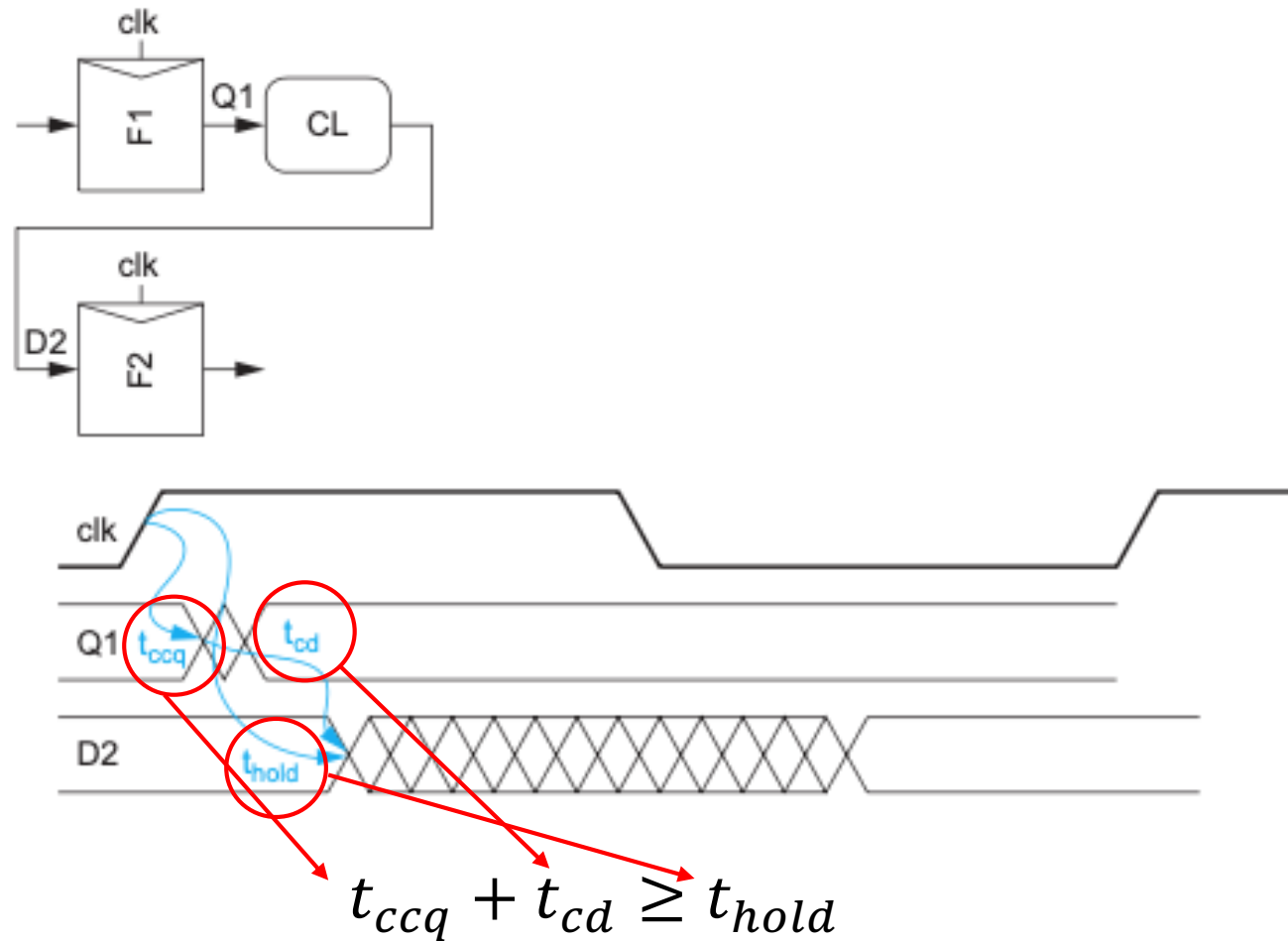
W&H p382

Pulsed Latch Maximum Delay Constraint



W&H p382-383

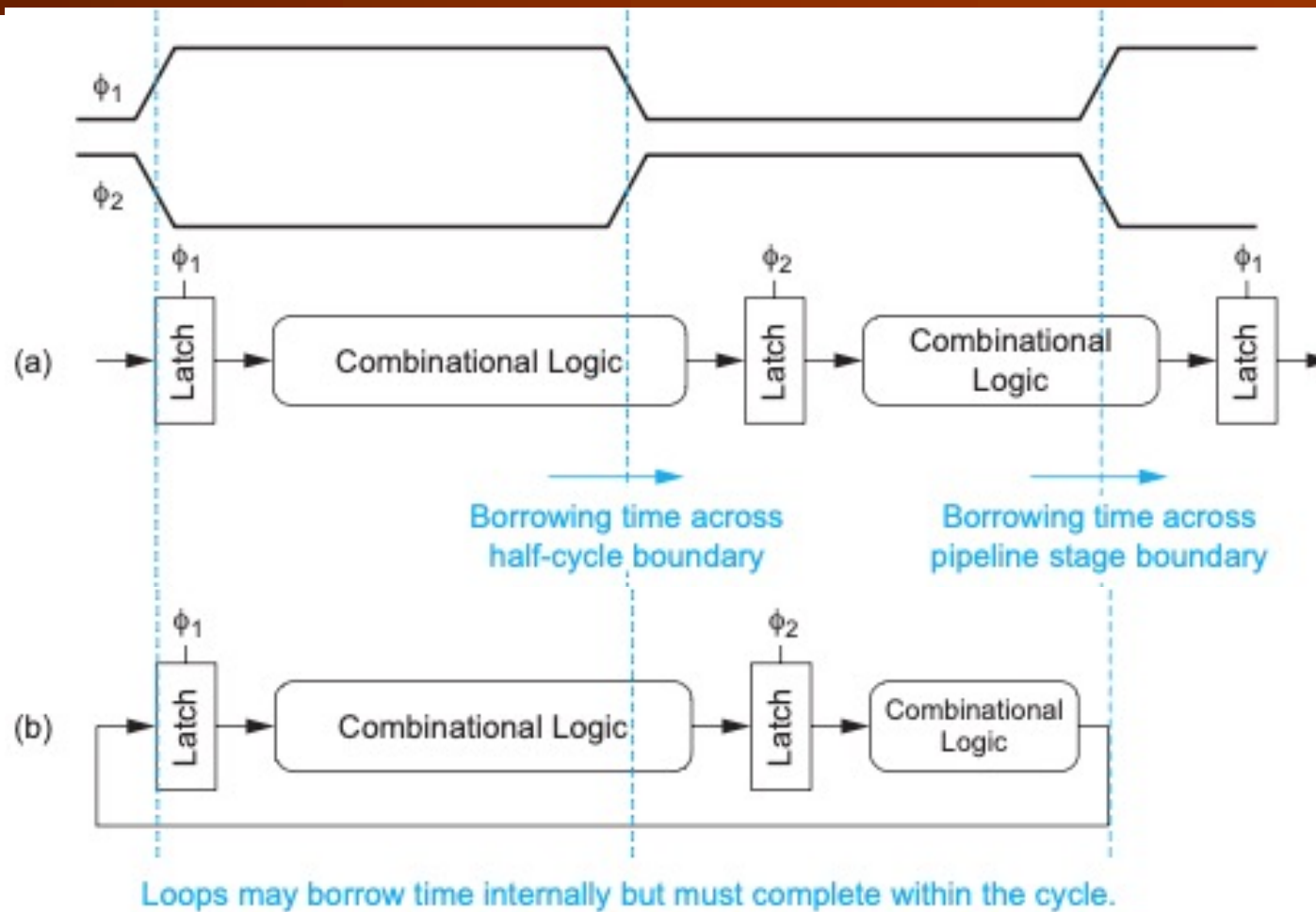
Flip-flop Minimum Delay Constraint



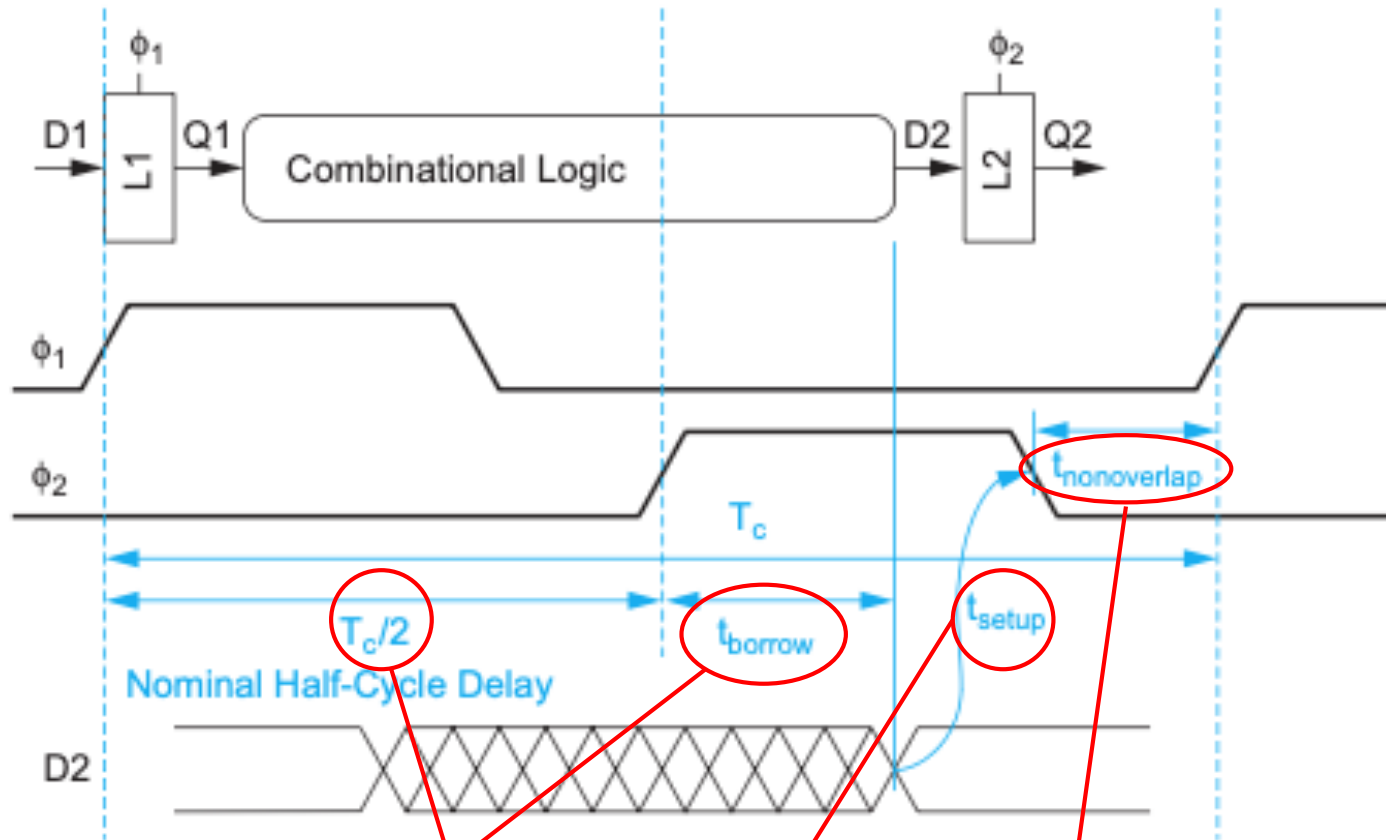
$$t_{cd} \geq t_{hold} - t_{ccq}$$

W&H p384-386

Time Borrowing (1)



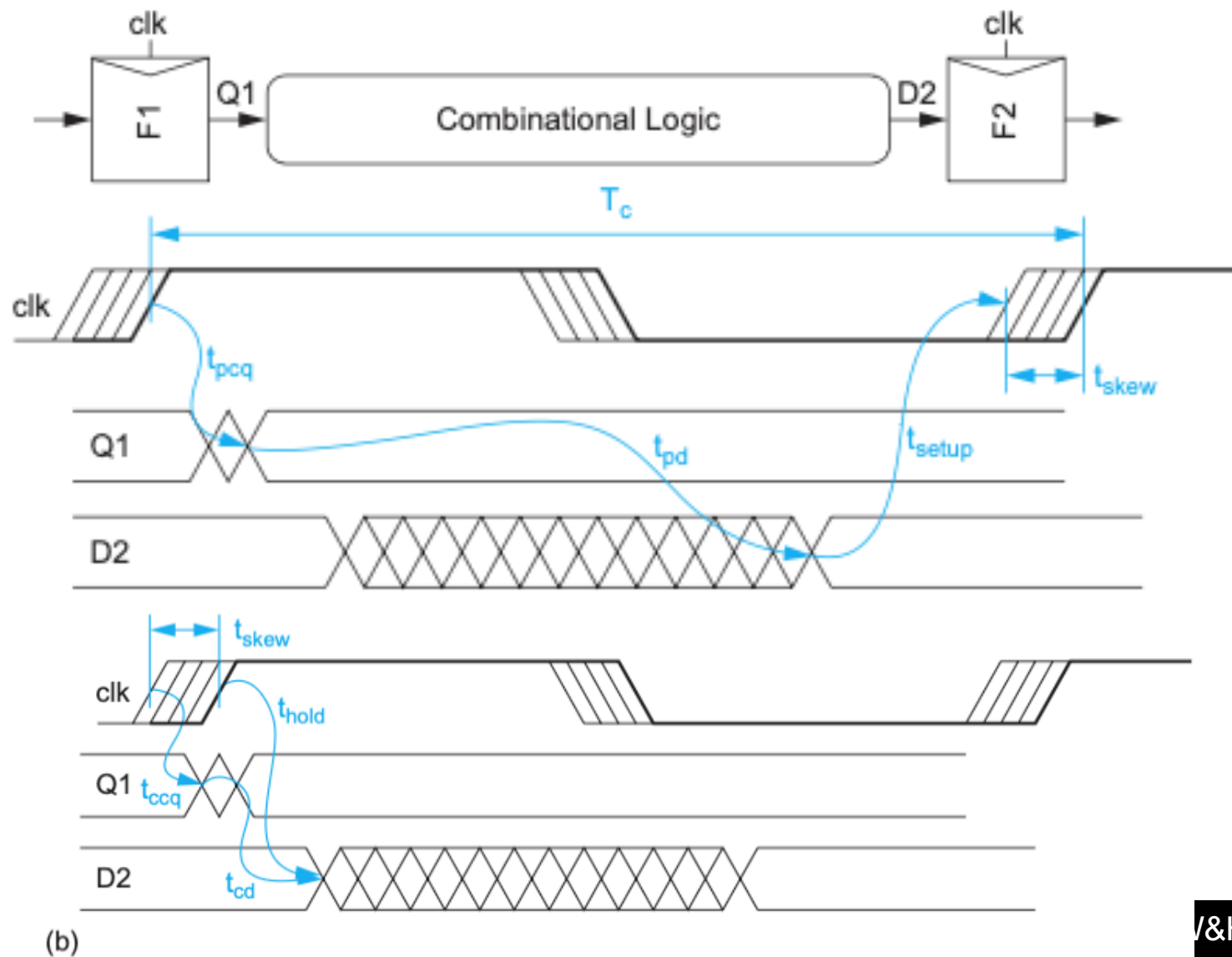
Time Borrowing (2)



$$t_{borrow} \leq \frac{T_c}{2} - (t_{setup} + t_{nonoverlap})$$

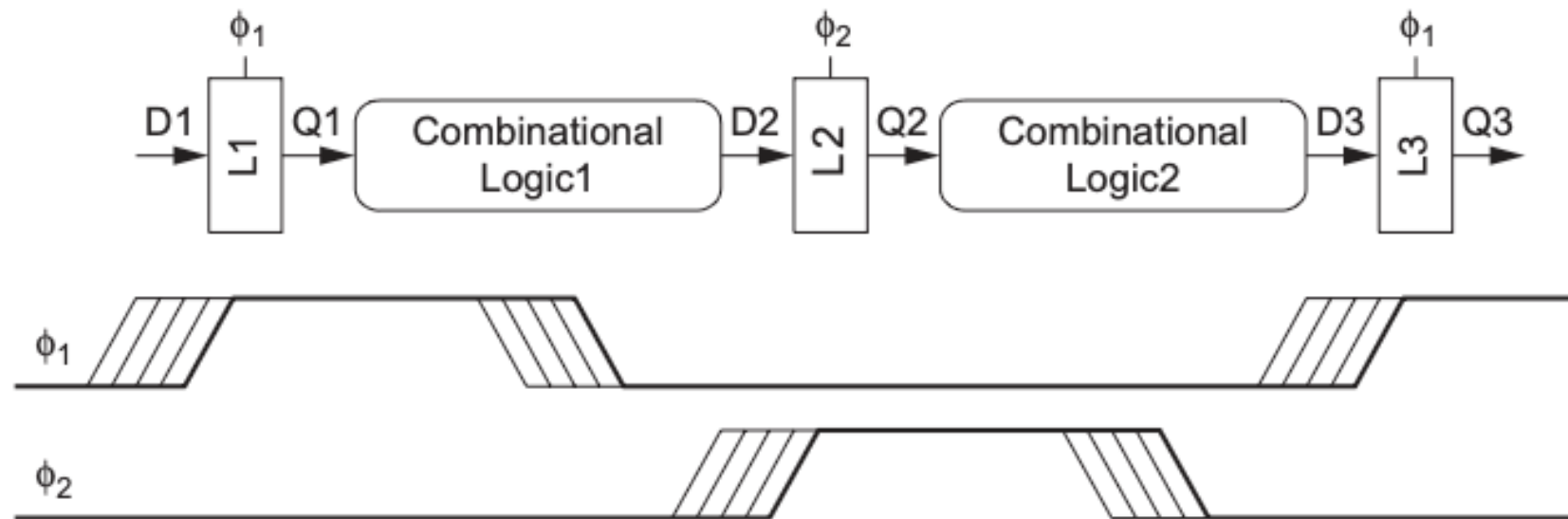
W&H 387-389

Clock Skew Problem in Flip-flop



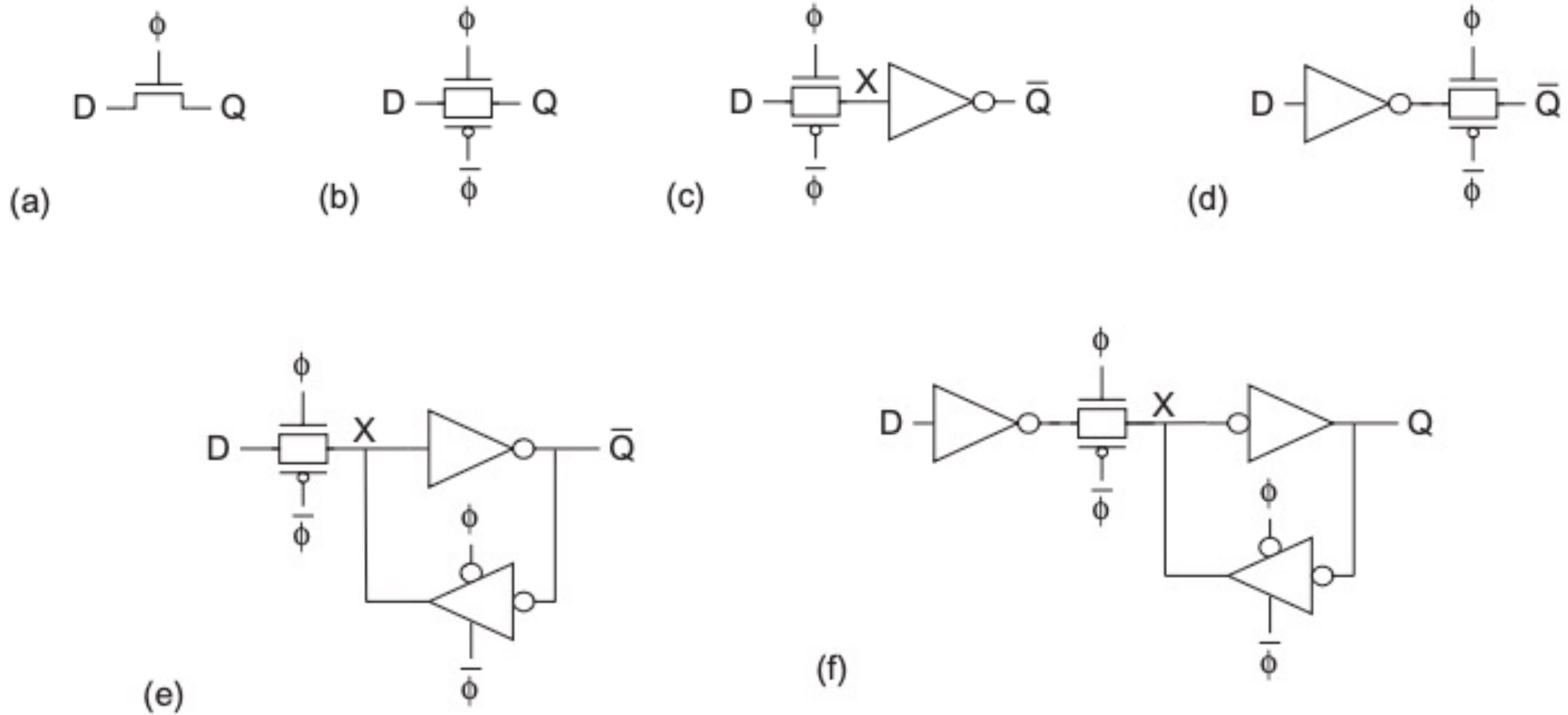
/&H 387-389

Clock Skew Problem in Transparent Latch



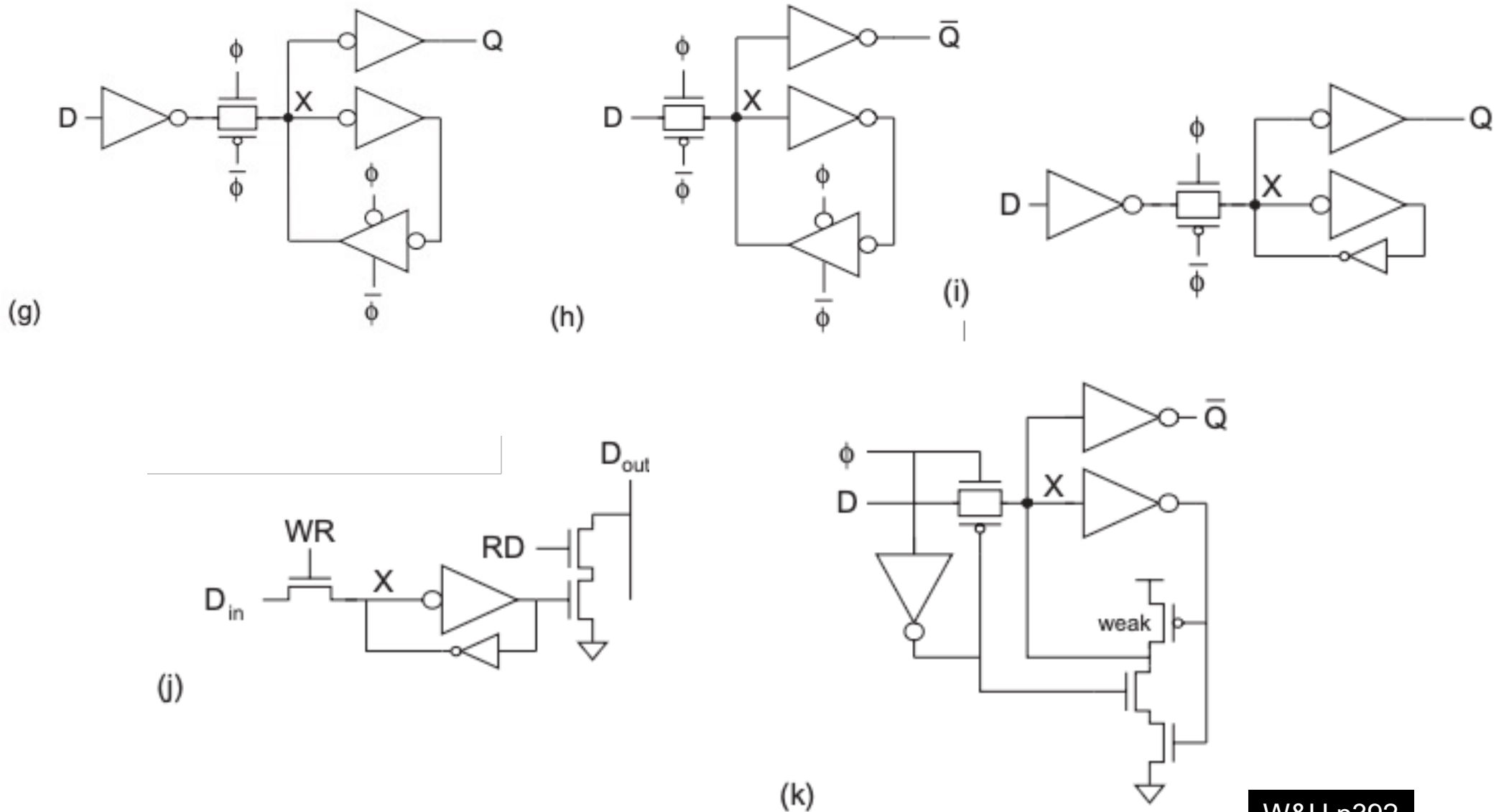
□ Transparent latches are **skew tolerant**.

Transparent Latch Circuits (1)



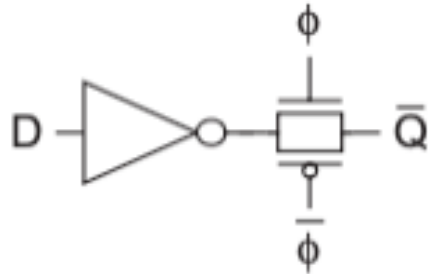
W&H p392

Transparent Latch Circuits (2)

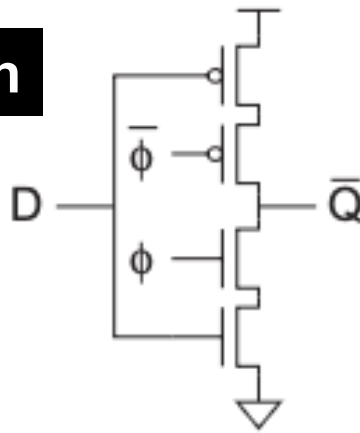
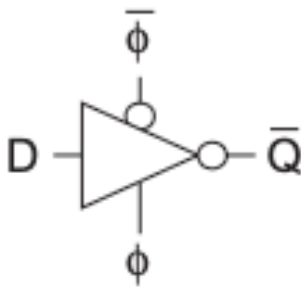


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C²MOS Latch Circuit

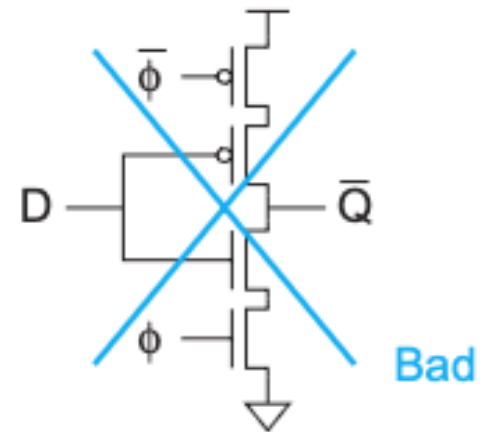


C²MOS Latch



(a)

❑ Charge sharing problem.

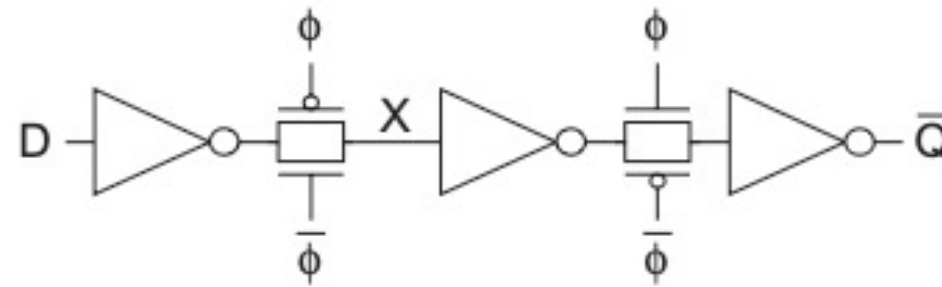


(b)

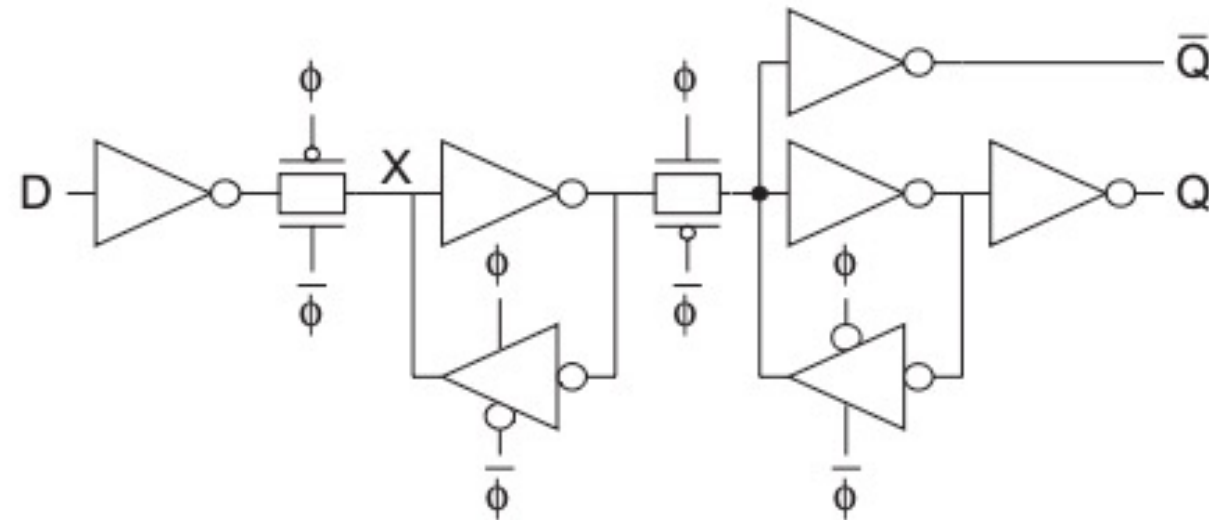
❑ Clocked CMOS Latch = a tri-state inverter

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Simple Flip-flop – single phase clock



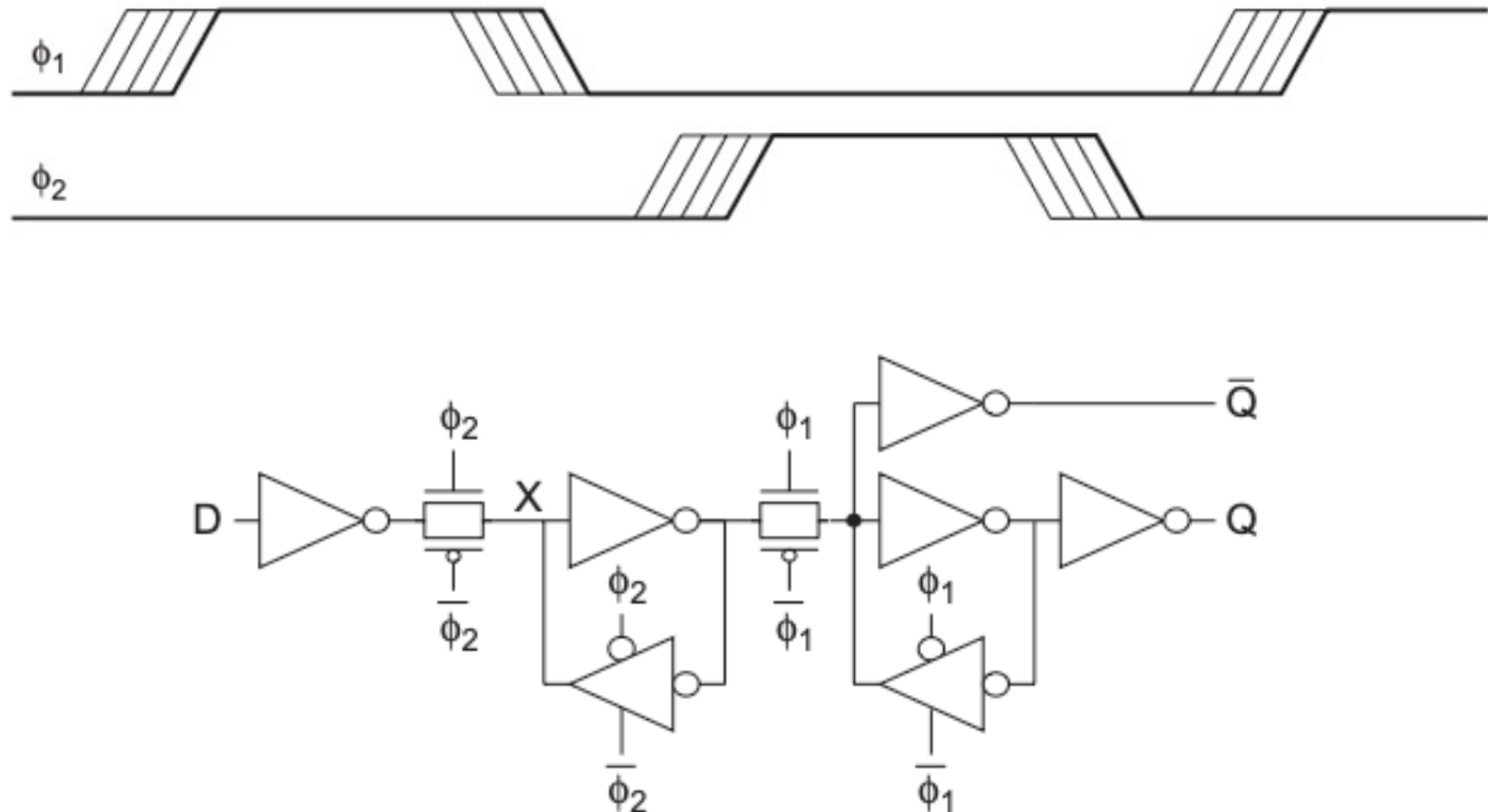
(a)



(b)

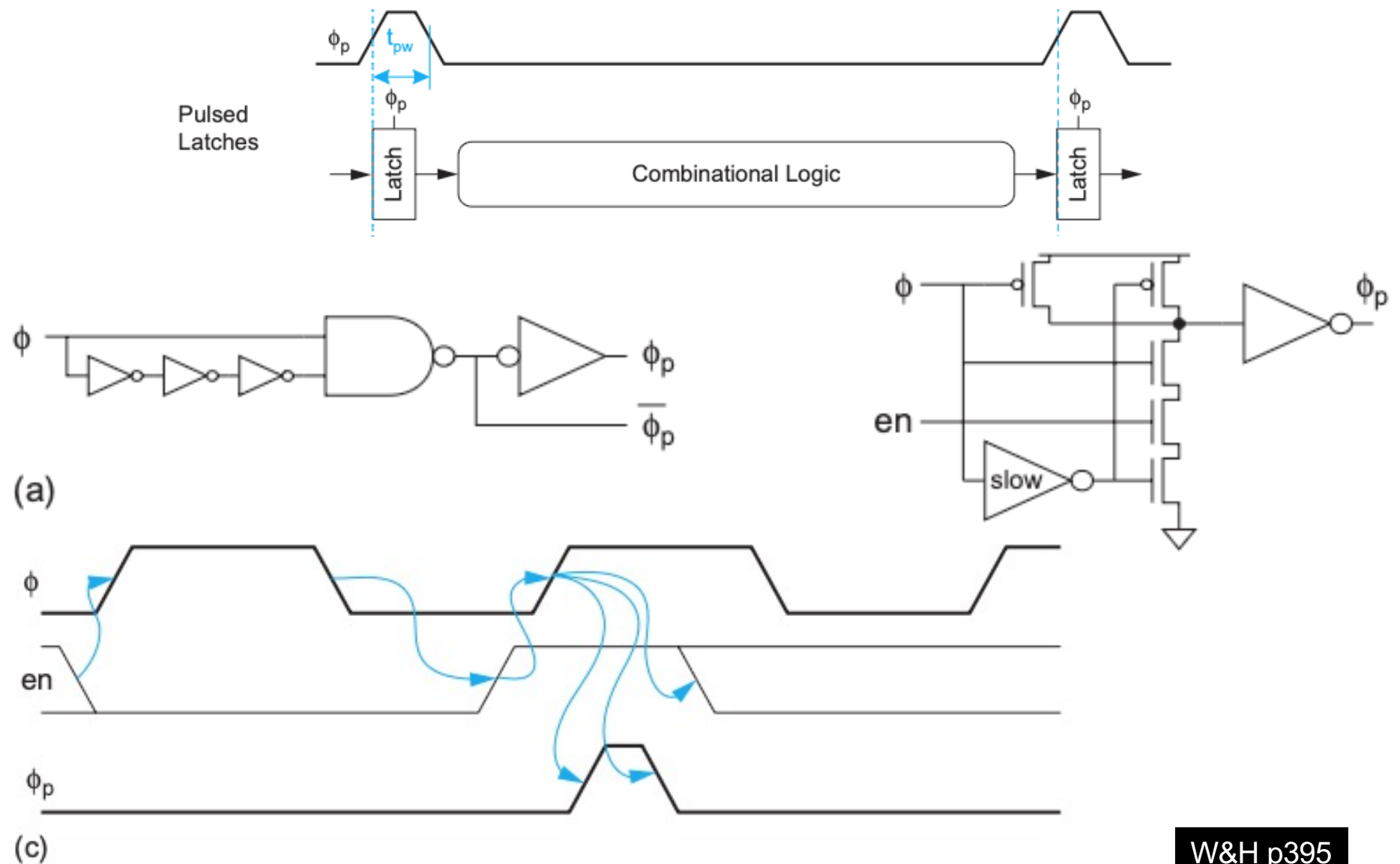
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Two-phase clocking Flip-flop



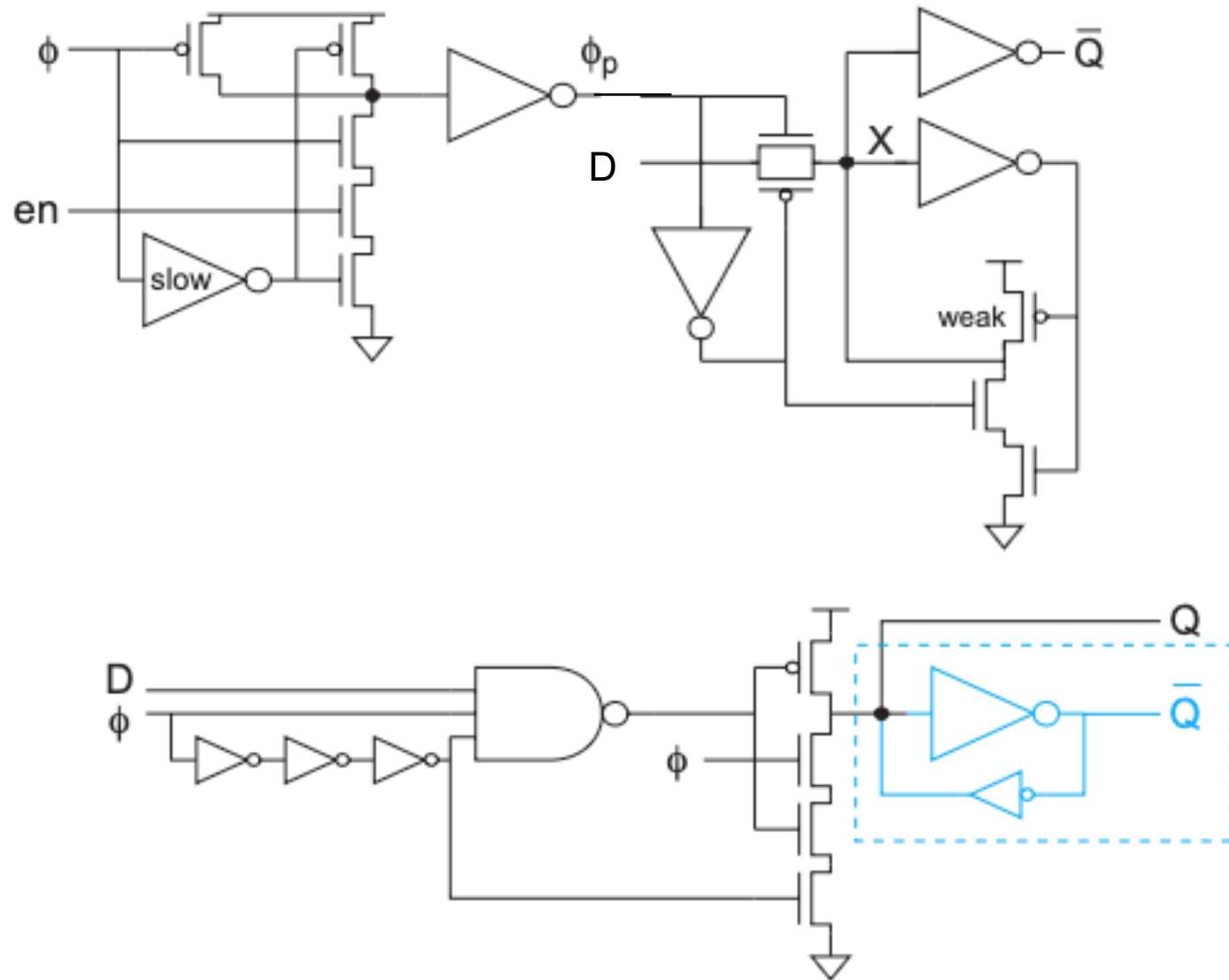
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Pulse Generators



W&H p395

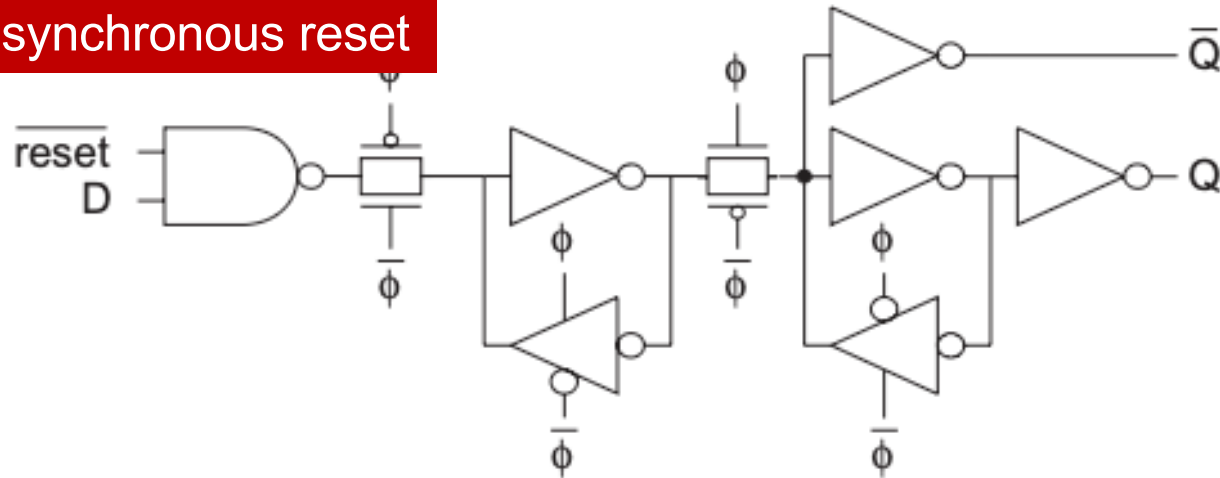
Pulse Latch Circuits



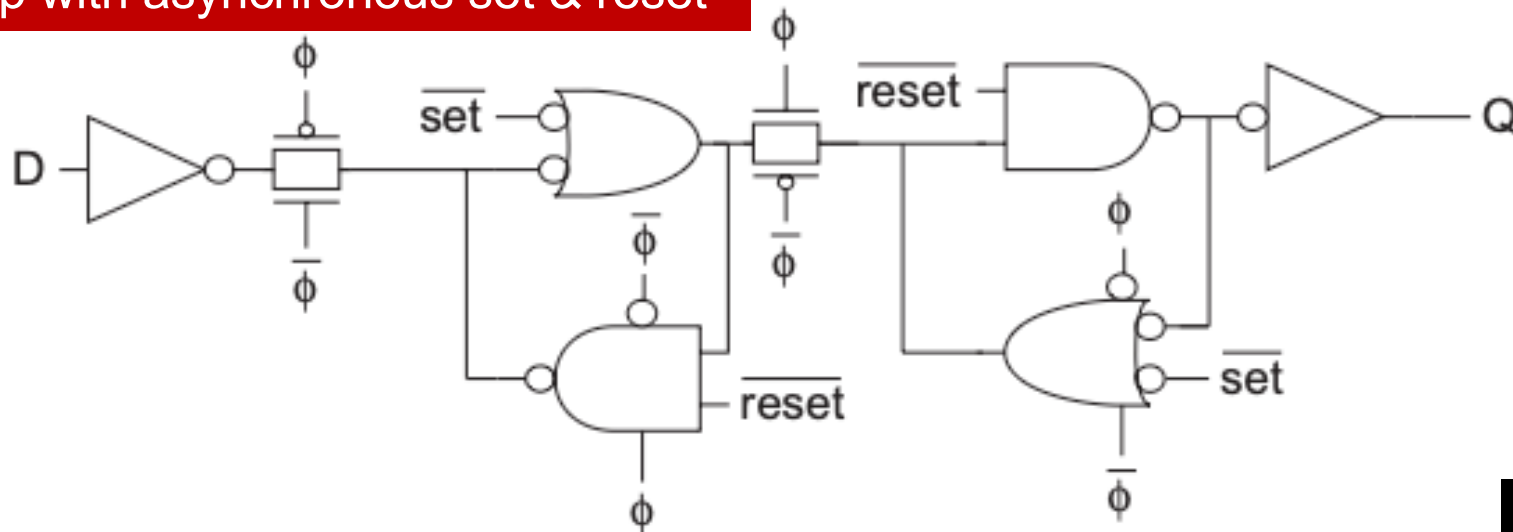
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Reset Flip-flop & Latches

Flip-flop with synchronous reset



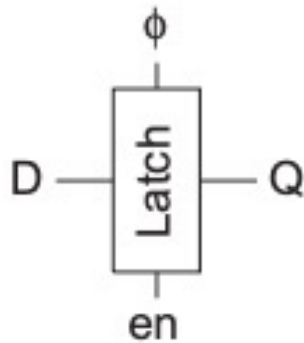
Flip-flop with asynchronous set & reset



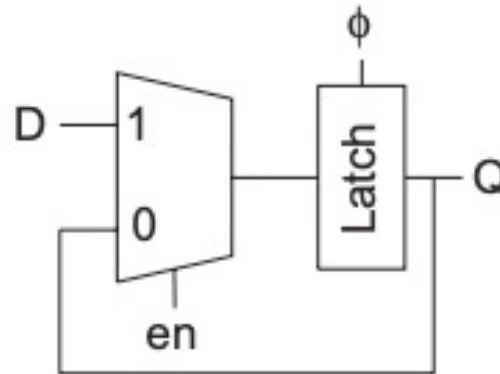
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Flip-flops & Latches with enable

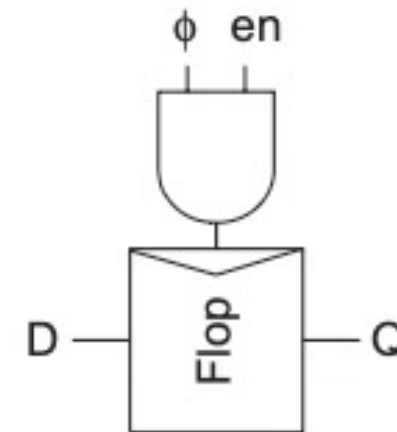
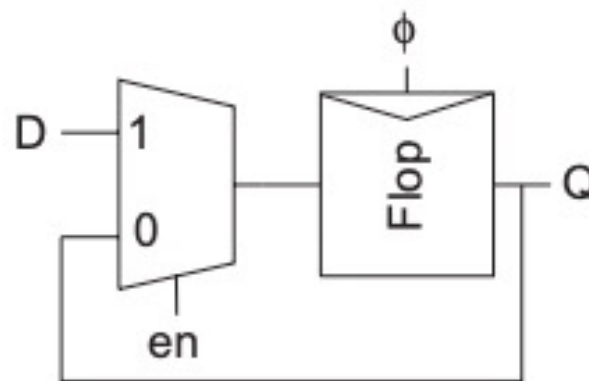
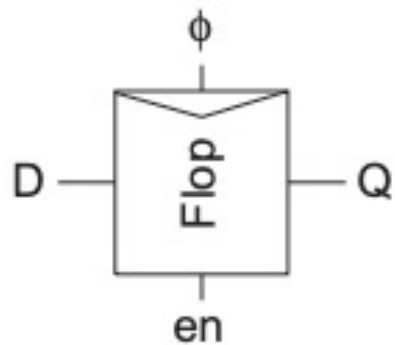
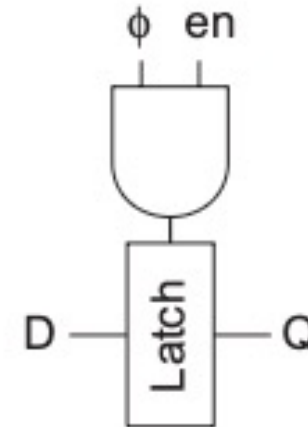
Symbol



Multiplexer Design



Clock Gating Design



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Advantage of Latches Over Flip-Flops

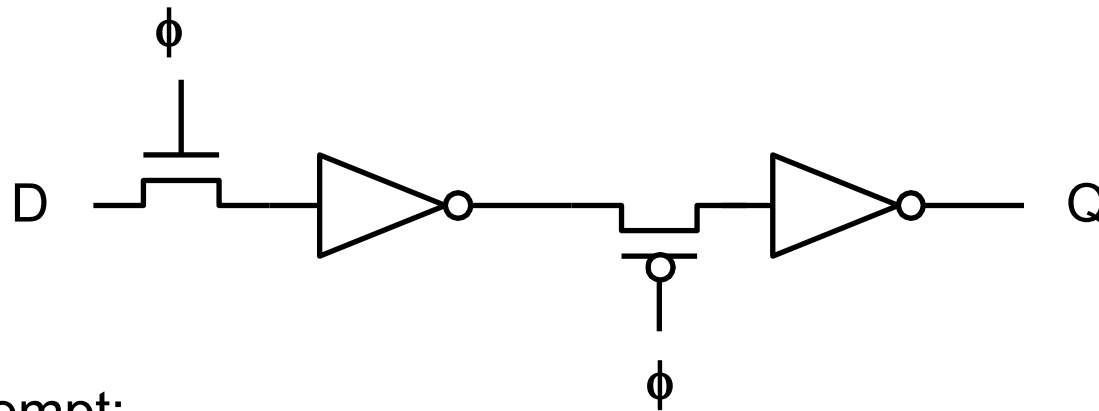
- ◆ If you are going to use clk and clk_b and control skew, why not go back to flip-flops?
- ◆ Many people do:
 - Most designs in industry are based on flip-flops
 - Very easy to verify timing
 - Each path between flip-flops must be less than cycle time
 - Tools check for skew and hold time violations
 - Short paths are padded (buffers are added to slow down the signals)
 - Skew in flip-flop based systems affects the critical path
- ◆ Latch designs are more flexible than a flip-flop design
 - Gives the designer more rope
 - Need to CAD tools to make sure it works
 - Can borrow time to allow a path to be longer than clock period
 - Can tolerate clock skew -- skew does not directly add to cycle time

Disadvantages of two-phase clocking

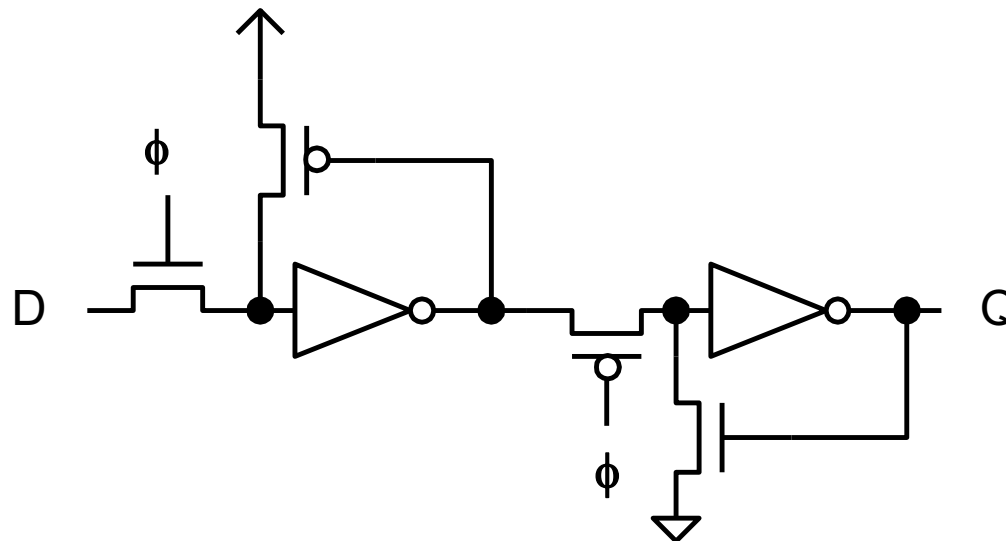
- ◆ Need four clocks in general
 - Need true and complement of both clocks
- ◆ Still need low skew for good performance
 - The skew increases the cycle time of the machine
 - Need low skew between all the clocks for good performance
 - Want to have $\Phi 1$ and $\Phi 2$ close to coincident
- ◆ Many systems use clock and its complement instead of 2 phases
 - Needless to say they are very careful about clock skew
 - For these systems it is still useful to maintain 2 phase timing types, since it ensures you connect all logic to the right latches
 - Call $\text{Clk} - \Phi 1$ and $\text{Clk} - \Phi 2$, and go from there.

μ - π latches: Poor man's TSPC Latch

- ◆ What is wrong with this TSPC Latch?



- ◆ Second attempt:



μ - π latches Final solution

